
Penultimate-Layer Feature-Based OOD Detection: Feature Shaping and Its Limits

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Abstract

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1 Introduction

Feature shaping refers to a family of post-hoc OOD detection methods that act on the penultimate-layer feature representation of a frozen, pre-trained classifier. Rather than reading the network’s output directly, these methods insert a fixed transformation that rewrites the feature vector before it reaches the final linear layer, so that the resulting OOD score separates in-distribution (ID) from out-of-distribution (OOD) inputs more sharply. The transformation is applied at test time only and is designed to leave ID classification essentially unchanged; no weights are retrained. Methods in this family differ in whether they reshape each feature coordinate independently (*element-wise*) or transform the feature vector jointly (*whole-vector*), and in how the reshaping rule is obtained.

ReAct [4] is the simplest: it caps each feature at an upper threshold, attenuating the abnormally large activations that OOD inputs tend to trigger. ASH [1] instead operates on the whole vector at once, keeping only a small fraction of the largest activations per input and discarding the rest before rescaling what remains; unlike ReAct, it draws on no statistics from training data. VRA [5] returns to per-coordinate reshaping but motivates its rule from a variational argument, suppressing both unusually small and unusually large activations while amplifying mid-range ones.

Optimal-FS [?] approaches the same goal from an optimization standpoint. Rather than proposing another hand-designed rule, it asks which feature values contribute most to distinguishing ID from OOD inputs, and derives the reshaping rule that best exploits them—obtainable in closed form from ID data alone. This optimization view doubles as a unifying account of the element-wise methods that came before it: ReAct and VRA can be recovered as approximations to its optimal element-wise shaping function. The account is, however, specific to element-wise shaping; whole-vector methods such as ASH lie outside the concrete problem it solves—a gap its authors leave open, and one this report revisits when asking what feature shaping is really exploiting.

The fifth method, BLOOD [2], is included as a deliberate contrast rather than another shaping rule. It leaves the feature vector untouched and instead measures how smoothly the network transforms its internal representations, on the premise that a trained model processes ID inputs more smoothly than OOD ones. Because it reads a property of the representation rather than editing it, BLOOD isolates the assumption the four shaping methods share—that the OOD signal resides in the penultimate

feature vector itself—and lets the report test whether that shared premise, rather than any particular reshaping recipe, is what does the work.

2 Related Work

Lu et al. provide a categorization of OOD methods that this paper will follow. According to their taxonomy, feature based OOD detection methods can be differentiated from other approaches by being training agnostic and post-hoc. This categorization contains all approaches that neither need to be implemented in a given model during training nor influence the training process of a given model. Feature based approaches are further categorized as post-hoc. This attribute is assigned to methods that can be applied to pretrained models at inference time. They therefore do not require to update model weights and biases. Lu et al. differentiate the OOD detection approaches further into feature based, gradient based, output based . . .

2.1 Positioning within the Survey Taxonomy

We adopt the mechanism-oriented taxonomy of Lu et al. [3] as the backbone for situating the methods studied in this report. Its top-level criterion is whether a method is permitted to intervene in the training process. *Training-driven* approaches improve OOD separability by modifying the optimization objective or the training data, whereas *training-agnostic* approaches operate on an already-trained, frozen model. The cluster examined here lies entirely on the training-agnostic side: none of the methods alter network weights or the training procedure.

Within training-agnostic detection, Lu et al. [3] draw a second distinction according to whether a method exploits statistical dependencies across the test stream. *Test-time adaptive* methods update the model or its running statistics from batches of incoming test data, while *post-hoc* methods score each input independently, reusing only quantities the frozen network already computes during a single forward pass. Their lightweight, deployment-friendly character makes post-hoc detection the dominant practical setting, and it is the setting to which every method discussed in this report belongs.

Post-hoc detectors are separated further by which model-internal quantity their scoring function reads: the output/logit vector (e.g. MSP, MaxLogit, Energy), a distance in feature space (e.g. Mahalanobis), an input or parameter gradient, the penultimate feature representation itself, or an estimated density. This report concentrates on the *feature-based* family, which derives its OOD signal from the penultimate-layer activation vector $\mathbf{z} \in \mathbb{R}^M$ produced before the linear classifier $\mathbf{W} \in \mathbb{R}^{M \times C}$.

Lu et al. [3] characterise feature-based methods as differing chiefly in whether they reshape activations directly, exploit a statistical property of the representation, or design a new metric over it. The first branch—*feature shaping*—replaces $\mathbf{z}$ with a transformed $\bar{\mathbf{z}} = f(\mathbf{z})$, $f : \mathbb{R}^M \rightarrow \mathbb{R}^M$, before the logits are recomputed, with f chosen so that the resulting score sharpens the ID/OOD gap while leaving ID classification largely intact. ReAct [4], ASH [1], VRA [5], and OptFS [6] form a progression within this branch, moving from heuristic intervention—activation truncation, top- K pruning with rescaling, and variationally motivated suppression and amplification—toward Optimal-FS, which casts shaping as an optimization problem and shows that the earlier methods implicitly approximate a single optimal, ID-only shaping function.

BLOOD [2] also reads the penultimate representation but does not reshape it. Instead it measures a statistical property of the network’s intermediate transformation—its smoothness, estimated through the feature Jacobian—and is therefore grouped by Lu et al. [3] with metric-design methods rather than with shaping. In this report it serves as a controlled counterpoint that isolates the shaping cluster’s shared assumption: that the discriminative OOD signal resides in the penultimate activation vector itself, recoverable by a value-to-value transformation of $\mathbf{z}$.

2.2 A Second Axis: Access Requirements

The mechanism taxonomy of Section 2.1 groups methods by *which* model-internal quantity they read. Jelenić et al. [2] argue that a second, arguably more practically relevant, axis groups methods by the *resources* they require, and partition detectors into three categories. *Black-box* methods need only input–output access and are therefore applicable even to models served behind a product API; the output-based scores MSP, MaxLogit, and Energy are the canonical examples. *White-box*

methods additionally require the model’s weights and architecture but no data, making them directly applicable to third-party pre-trained checkpoints. *Open-box* methods carry the heaviest prerequisites: unrestricted access permitting intervention in training, or use of the training set or a separate OOD validation set.

This axis is orthogonal to the mechanism taxonomy and cuts across the feature-based family studied here. Although ReAct [4], ASH [1], VRA [5], and Optimal-FS [6] share a shaping mechanism, they differ in access requirements: ASH is purely data-free—its top- K pruning needs only the weights, placing it on the white-box side—whereas ReAct’s percentile threshold, VRA’s calibrated cut-offs, and Optimal-FS’s closed-form shaping function are all fit to ID data, placing them in the open-box category. BLOOD [2] reads a different quantity (transformation smoothness via the feature Jacobian) but, requiring only the weights, is likewise white-box. The clean contrast on this axis is therefore not BLOOD against the shaping cluster, but the data-free methods (ASH and BLOOD) against the ID-data-dependent ones (ReAct, VRA, Optimal-FS).

The open-box category also supplies the strongest reference point. Mahalanobis distance [?], which fits class-conditional Gaussians to ID training features and is thus open-box, serves throughout this report as an honest ceiling rather than a member of either shaping or smoothness paradigm. In the white-box comparisons of Jelenić et al. [2] it outperforms BLOOD in nearly every transformer setting, and across benchmarks it frequently exceeds both the shaping cluster and BLOOD—so neither feature-based paradigm is the overall top performer. This dominance is, however, setting-specific: Mahalanobis is strong on transformer/text backbones but markedly weaker on ImageNet-scale convolutional networks, where its class-conditional Gaussian assumption.

3 Methodology

We consider a classifier pre-trained on in-distribution (ID) data and held frozen. For an input $\mathbf{x}$, the backbone produces a penultimate feature vector $\mathbf{z} \in \mathbb{R}^M$, which a linear head $\mathbf{W}$ maps to C class logits. A post-hoc detector reduces the network’s response to a scalar score $s(\mathbf{x}) \in \mathbb{R}$ and flags the sample once that score crosses a threshold. Detection is thus a thresholding rule

$$G_\lambda : \mathbb{R} \rightarrow \{\text{ID}, \text{OOD}\}, \quad (1)$$

parameterized by a threshold λ , that labels $\mathbf{x}$ by comparing $s(\mathbf{x})$ to λ . The rule G_λ is common to every method; the methods differ only in how the score s is computed from the frozen network and in its orientation (which side of λ denotes ID). Detection quality is reported with the two standard metrics, AUROC (threshold-free separability) and FPR95 (false positive rate at 95% true positive rate).

3.1 Feature Shaping

Feature shaping inserts a transformation $f : \mathbb{R}^M \rightarrow \mathbb{R}^M$ at the penultimate layer, reshaping $\mathbf{z}$ before the logits are recomputed and scored. The detector then factors into a shaping stage and a scoring stage,

$$f : \mathbb{R}^M \rightarrow \mathbb{R}^M, \quad s : \mathbb{R}^C \rightarrow \mathbb{R}, \quad s_{\text{shaped}}(\mathbf{x}) = s(\mathbf{W}f(\mathbf{z})), \quad (2)$$

and the sample is flagged by $G_\lambda(s_{\text{shaped}}(\mathbf{x}))$. The score s is a standard logit-based score (MSP, MaxLogit, or energy), for which a lower value indicates OOD; crucially, the choice of s is orthogonal to the shaping function f , a separation Optimal-FS later exploits. Setting $f = \text{id}$ recovers the unshaped baseline. Shaping functions are *element-wise* when f acts coordinatewise, $\bar{z}_i = f(z_i)$, and *whole-vector* when the transformation of one coordinate depends on the others; this distinction, carried entirely by the single symbol f , separates the methods below.

3.2 ReAct and VRA

ReAct [4] and VRA [5] are both element-wise. ReAct clips each activation at an upper threshold c ,

$$f_{\text{ReAct}}(z_i) = \min(z_i, c), \quad (3)$$

with c set to a high percentile (typically the 90th) of the activations estimated on ID data. The motivation is that OOD inputs tend to produce a few abnormally large activations; capping them

suppresses the OOD score more than the ID score while leaving most ID activations, which fall below c , untouched.

VRA starts from a variational analysis of which operation maximally separates ID and OOD activations, concluding that the ideal rule suppresses both abnormally low and abnormally high activations. Its canonical piecewise realization, VRA-P, uses two ID-estimated quantile thresholds $\alpha < \beta$,

$$f_{\text{VRA-P}}(z_i) = \begin{cases} 0 & z_i < \alpha, \\ z_i & \alpha \leq z_i \leq \beta, \\ \beta & z_i > \beta. \end{cases} \quad (4)$$

This strictly generalizes ReAct, recovered as the special case $\alpha = 0$: VRA-P adds a lower cut that zeroes the abnormally small activations ReAct leaves in place. Both methods fit their thresholds to ID data and are therefore data-dependent (cf. Section 2.2). The amplification of mid-range activations (VRA+) and a feature–logit combined score (VRA++) are further refinements that depart from pure shaping; we summarize them in the appendix.

3.3 ASH

ASH [1] operates on the whole vector. For each sample it computes the p th percentile t of that sample’s own activations, prunes everything below t to zero, and then treats the surviving activations in one of three ways: ASH-P leaves them unchanged, ASH-B sets them all to a common constant, and ASH-S—the version we take as canonical—scales the survivors by a factor that restores the pre-pruning activation mass,

$$f_{\text{ASH-S}}(z_i) = \begin{cases} z_i \cdot \exp(s_1/s_2) & z_i \geq t, \\ 0 & z_i < t, \end{cases} \quad s_1 = \sum_j z_j, \quad s_2 = \sum_{j: z_j \geq t} z_j, \quad (5)$$

where s_1 and s_2 are the sums of activations before and after pruning. The premise is that the representation is heavily redundant, so most of it can be discarded and the sparse surviving support—rescaled to compensate—still classifies ID inputs while sharpening OOD separation. Because the threshold t is computed per sample from the whole vector, ASH is whole-vector and not reducible to an element-wise map, and it uses no training statistics, making it data-free (cf. Section 2.2). An extreme ablation, ASH-Rand, replaces the surviving values with random ones; we return to it in the discussion, as it bears on what ASH actually exploits.

3.4 Optimal-FS

Optimal-FS [?] recasts element-wise shaping as an optimization problem. Writing the per-coordinate reshaping as a multiplicative factor $\theta(z_i) = f(z_i)/z_i$, the shaped maximum logit is

$$\bar{\ell}_{\max}(\bar{\mathbf{z}}) = \sum_{i=1}^M \theta(z_i) w_i^{\max} z_i, \quad (6)$$

so the question becomes which feature *value ranges* should be up- or down-weighted. Optimal-FS partitions the range of feature values into K intervals $\{[a_k, b_k]\}_{k=1}^K$, treats θ as a piecewise constant $\hat{\theta}_k$ over each, and collects the per-interval contributions into a K -dimensional interval-specific feature impact (ISFI) vector $\mathbf{I}(\mathbf{z})$, giving $\bar{\ell}_{\max} = \boldsymbol{\theta}^\top \mathbf{I}(\mathbf{z})$. The shaping vector is then optimized to separate the ID and OOD maximum logits, under a norm constraint that prevents trivial rescaling,

$$\max_{\boldsymbol{\theta}} \boldsymbol{\theta}^\top (\mathbb{E}_{\text{ID}}[\mathbf{I}(\mathbf{z})] - \mathbb{E}_{\text{OOD}}[\mathbf{I}(\mathbf{z})]), \quad \text{s.t. } \|\boldsymbol{\theta}\| = \sqrt{K}. \quad (7)$$

Solved with OOD data, this $\boldsymbol{\theta}$ closely matches the element-wise rules above, recovering them as approximations to a common optimum. Arguing that the OOD term contributes little beyond the ID direction, Optimal-FS drops it and obtains a closed-form solution from ID data alone,

$$\boldsymbol{\theta}^* = \frac{\sqrt{K}}{\|\mathbb{E}_{\text{ID}}[\mathbf{I}(\mathbf{z})]\|} \mathbb{E}_{\text{ID}}[\mathbf{I}(\mathbf{z})]. \quad (8)$$

Nearly all of the benefit comes from $K \leq 10$, indicating the optimal shaping function has a simple form (the full ISFI derivation is in the appendix). Two caveats matter for this report. First, the optimization is specific to element-wise shaping, so it unifies ReAct, VRA-P, and BFact but explicitly *does not* capture whole-vector methods such as ASH, which the authors name as an open problem. Second, like ReAct and VRA, it depends on ID data (cf. Section 2.2).

3.5 BLOOD

BLOOD [2] reads a different signal from the same region of the network. Rather than reshaping $\mathbf{z}$, it measures how smoothly the network transforms its representations, on the premise that a trained model maps ID inputs through smoother transformations than OOD ones. Smoothness at layer l is the squared Frobenius norm of that layer’s Jacobian $\mathbf{J}_l$,

$$\phi_l(\mathbf{x}) = \|\mathbf{J}_l(\mathbf{h}_l)\|_F^2, \quad (9)$$

estimated without forming the Jacobian via N pairs of isotropic random probes $\mathbf{v}_i, \mathbf{w}_i$ (zero mean, unit variance),

$$\hat{\phi}_l(\mathbf{x}) = \frac{1}{N} \sum_{i=1}^N (\mathbf{w}_i^\top \mathbf{J}_l(\mathbf{h}_l) \mathbf{v}_i)^2, \quad \mathbb{E}[\hat{\phi}_l(\mathbf{x})] = \|\mathbf{J}_l\|_F^2, \quad (10)$$

the isotropy of the probes being what makes the estimator unbiased. BLOOD uses this quantity directly as its score, $s(\mathbf{x}) = \hat{\phi}_{L-1}(\mathbf{x})$, with the *inverted* orientation relative to the shaping cluster: a higher value indicates OOD. The detector G_λ still applies, only with the sides of λ reversed. Note that this score is not expressible in the shaping composition $s \circ \mathbf{W} \circ f$ of Section 3.1: it reads the Jacobian and never factors through the head or a shaping map, which is precisely why BLOOD sits outside feature shaping. The method has two variants, BLOOD_M averaging $\hat{\phi}_l$ over all layers and BLOOD_L using only the final transformation, the Jacobian of the head with respect to the penultimate representation $\mathbf{z}$. The authors focus on BLOOD_L, which outperforms BLOOD_M more often, attributing this to the final layer undergoing the most training-driven change; this coincides with our penultimate-layer scope, so we take BLOOD_L as canonical. BLOOD requires only the model’s weights and no training data (cf. Section 2.2).

4 Experiments

The five methods were never evaluated under a shared protocol, and no honest unified leaderboard exists: they differ in modality (vision vs. text), backbone, OOD-dataset suite, and scoring function. Rather than force a comparison the literature does not support, we read the evidence as two internally consistent *islands*—vision feature shaping and text between-layer smoothness—and treat their incomparability as itself a finding: these methods target different deployment settings, and their relative merits are only defined within a setting. All figures below are reproduced from the cited papers, not independently re-run, and are comparable only within an island.

4.1 Vision: the shaping cluster under one protocol

Optimal-FS [?] provides the only head-to-head evaluation of the shaping cluster, re-running ReAct, ASH (P/B/S), VRA-P, and its own method across eight ImageNet-1k backbones under one protocol. On ResNet-50—the architecture all of these methods were designed on—the shaping methods are strong and close together (AUROC ≈ 90.3 for ASH-S, 88.6 for VRA-P, 86.5 for ReAct, 88.6 for Optimal-FS). The picture inverts off ResNet-50. On transformer and MLP backbones the most aggressive shaping rules collapse toward chance: ASH-S falls to 18.5 AUROC on ViT-B-16 and 33.9 on MLP-B, and VRA-P behaves similarly, dragging their eight-backbone averages to ≈ 38 . ReAct degrades far more gracefully (averaging 81.3, never falling below the high 70s), and Optimal-FS holds across the board (83.0 average, 82.7 on ViT-B-16). So the cluster’s apparent unity is conditional on its home architecture; the clip-based ReAct and the ID-optimized Optimal-FS generalize, while the percentile-prune methods do not. On the smaller-scale CIFAR benchmarks, feature shaping as a whole underperforms simple logit baselines, with Optimal-FS competitive but not dominant. Two caveats apply: these numbers come from a paper with a stake in the comparison, and the eight-dataset ImageNet averaging differs from the four-dataset suites used in the original ASH and ReAct papers.

4.2 Text: between-layer smoothness

BLOOD [2] evaluates a disjoint setting—text classification with RoBERTa and ELECTRA over eight ID datasets—so it neither confirms nor contradicts the vision results. Among methods requiring no training data (white/black-box), BLOOD_L is the strongest on most datasets, and more consistently than the all-layer variant BLOOD_M; the authors accordingly focus on BLOOD_L, attributing its

edge to the final layer absorbing the most training-driven change. The advantage is clearest on ELECTRA, where BLOOD_L leads the white/black-box group on all eight datasets. Notably, the two shaping methods that appear here, ASH and ReAct, are only middling in this setting, indicating that vision-tuned shaping does not transfer to text. BLOOD’s own grouping also mirrors our access axis (Section 2.2): ASH is filed as white/black-box (data-free) and ReAct as open-box (it needs an ID-data percentile).

4.3 The ceiling, and what the islands show

A distance-based open-box method, Mahalanobis distance [?], bounds both islands from above. In BLOOD’s comparison it outperforms BLOOD in every setup but one (ELECTRA on TREC), and it is consistently the top scorer on the transformer text backbones. Its dominance is, however, setting-specific rather than universal: on ImageNet-scale ConvNets it is weak (AUROC ≈ 55 on ResNet-50 in VRA’s table [5]), where its class-conditional Gaussian assumption is harder to satisfy. The two islands therefore deliver a consistent structural message rather than a numerical ranking: feature shaping wins only on its home architecture and modality, between-layer smoothness occupies a different modality entirely, and a simple distance-based detector outperforms both wherever its distributional assumptions hold—so neither shaping nor smoothness is the better paradigm in any setting-free sense.

5 Discussion

The five methods share the penultimate layer of a frozen network but were developed separately, on different architectures and modalities, and never co-evaluated under one protocol. Placing them together raises three questions: why the same methods succeed on some architectures and fail on others (Section 5.1); how far the shaping cluster can be formally unified (Section 5.2); and whether the OOD signal lies in the feature values or in the transformation applied to them (Section 5.3).

5.1 Architecture dependence

Shaping rules designed on ResNet-50 do not transfer: ASH-S falls from 90.3 to 18.5 AUROC from ResNet-50 to ViT-B-16 [?]. Optimal-FS reports this as a pattern—for shaping methods, convolutional-network performance is negatively correlated with performance on other architectures [?]. The methods’ mechanisms account for it. ASH’s pruning assumes a redundant, sparse representation and is framed as removing excess features of an over-parameterized network [1], a ReLU convolutional premise. BLOOD instead reads transformation smoothness in the upper layers of a transformer, where fine-tuning concentrates and where ID and OOD representations diverge [2]. We read these together as one point: each method encodes the architecture it was designed on, which is why neither transfers.

ReAct gives a partial cause. It attributes the abnormal OOD activations to batch-normalization statistics estimated on ID data and applied to OOD inputs, making batch normalization the exploited signal rather than an obstacle [4]. ReAct also helps under weight and group normalization [4], so the effect is not specific to batch normalization. The architecture dependence is therefore real but its root cause is unresolved across the cluster.

5.2 How far the shaping cluster unifies

Optimal-FS defines shaping as optimization over functions $f : \mathbb{R}^M \rightarrow \mathbb{R}^M$ and proves that ReAct and VRA-P, rewritten via $\theta(z) = f(z)/z$, approximate the optimal element-wise function [?]. This optimum is simple: most of the benefit occurs at $K \leq 1$ intervals, which prunes only the tails of the feature distribution [?]. This explains why ReAct, a single clip, stays competitive with the optimization-derived method on the architectures both were tuned for: little structure beyond the tails remains to recover.

The unification is bounded, and Optimal-FS draws the boundary. Its tractable instance restricts to element-wise f and, in its own statement, fails to capture whole-vector methods such as ASH, which it still places inside the general $\mathbb{R}^M \rightarrow \mathbb{R}^M$ class [?]. ASH is thus excluded by the reduction, not the definition, and the whole-vector extension is named as open future work [?]. Extending this

boundary to a case Optimal-FS does not treat, we observe that BLOOD lies outside the class: its score is the Frobenius norm of a Jacobian [2], not a map returning a reshaped feature vector, so no extension of the optimization can absorb it. This gives three tiers: ReAct and VRA-P inside the solved problem, ASH inside the general class but outside the solved instance, and BLOOD outside the class. The first gap is one of tractability; the second is categorical.

Whether the whole-vector extension is reachable is open. Two findings bear on it, and connecting them is our reading rather than a stated result. The element-wise optimum is coarse [?], and ASH’s signal is also coarse: ASH-Rand, which replaces the surviving activations with random values, still beats the logit baselines, so the signal lies in the support and the global scale rather than the value profile [1]. We suggest both point the same way—a whole-vector optimum would also be simple, consistent with Optimal-FS’s conjecture [?]. The open obstacle is that such a formulation must optimize which coordinates survive, a combinatorial selection unlike the element-wise interval weights. The same finding qualifies the “prior methods approximate the optimum” reading for ASH: it is a curve-to-curve match for the element-wise methods [?], but for ASH the correspondence is to the support-and-scale operation, not a value curve.

5.3 Where the signal lives

Because BLOOD is not a shaping function, it serves as a counterpoint that differs from the cluster on one variable: what is measured at the top of the network. The four shaping methods edit the activation vector; BLOOD leaves it unchanged and measures the smoothness of the transformation applied to it [2]. This poses a question neither paradigm sets out to answer: is the OOD signal a property of the feature values or of the map applied to them?

We do not resolve which is fundamental. The juxtaposition does narrow the cluster’s premise. BLOOD shows the activation vector is not the only carrier of OOD signal at the penultimate layer, and ASH-Rand shows that within the cluster the fine values are not the operative part. The shared assumption—that the signal can be recovered by reshaping the penultimate activation vector—is therefore narrower than its framing, though not incorrect.

6 Conclusion

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Use of Large Language Models

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A Appendix

Table 1: Unified notation used throughout this report. Symbols are grouped by the work that introduces the corresponding quantity in the OOD-detection context. Typographic conventions—bold vectors and matrices, $\mathbb{1}\{\cdot\}$, and $\top$ for transpose—follow ?]. Where a symbol is renamed for cross-paper consistency, the original symbol is given in parentheses.

Symbol	Meaning
<i>General notation</i>	
x	Input sample
z	Penultimate-layer feature (activation) vector
z_i	i -th element of z
$\tilde{z}$	Feature vector after shaping / rectification
M	Penultimate feature dimension
C	Number of classes
$\ell_i, \ell^{\max}$	i -th logit; max logit
$\mathbf{J}$	Jacobian matrix
$\ \cdot\ _F$	Frobenius norm
$\mathbb{1}(\cdot)$	Indicator (1 if condition holds, else 0)
$\mathbb{E}[\cdot]$	Expectation
Sun et al. [4]	
c	Activation clip threshold, $\text{ReAct}(z; c) = \min(z, c)$
$S(\cdot)$	OOD scoring function (MSP, MaxLogit, Energy, ...)
λ	OOD decision threshold
G_λ	OOD detector (in/out rule on S)
Xu et al. [5]	
$p_{\text{in}}, p_{\text{out}}$	ID / OOD marginal densities over activations
$g(\cdot), g^*(\cdot)$	Rectification function; variational optimum
τ	Fidelity-gap trade-off coefficient (orig. λ)
α, β	Low / high activation thresholds
γ	VRA+ amplification offset on the intermediate band
η_α, η_β	ID quantiles defining α, β
Zhao et al. [6]	
$\mathbf{w}^{\max}$	Weight vector of the predicted (max-logit) class

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Symbol	Meaning
$\theta(\cdot)$	Element-wise shaping function, $\theta(z) := f(z)/z$
$\boldsymbol{\theta}$	Per-bucket shaping vector, $\boldsymbol{\theta} \in \mathbb{R}^K$
$I_{[a,b]}(\mathbf{z})$	Interval-specific feature impact (ISFI) of bucket $[a, b)$
a_k, b_k	Edges of the k -th bucket
K	Number of buckets ($K = 100$)
Jelenić et al. [2]	
$\phi_l(x)$	Score: squared Frobenius norm of the layer- l Jacobian
h_l	Layer- l representation ($h_{L-1} = \mathbf{z}$)
N	Number of random probes (orig. M)
$\mathbf{v}_i, \mathbf{w}_i$	Gaussian probe / projection vectors
$\mathbf{u}_i$	Jacobian-vector product, $\mathbf{u}_i = \mathbf{J}_{L-1} \mathbf{v}_i$
Djurisic et al. [1]	
p	Pruning percentile
t	Per-sample threshold, $t = \text{Perc}_p(\mathbf{z})$
m_i	Binary keep-mask, $m_i = \mathbb{1}\{z_i \geq t\}$
n	Number of surviving activations
s_1, s_2	Pre- / post-pruning activation sums